Dataset View - Custom

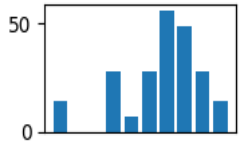
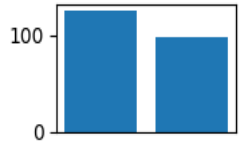
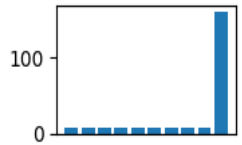
Dataset Head

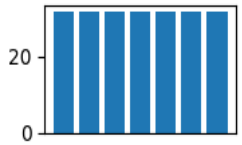
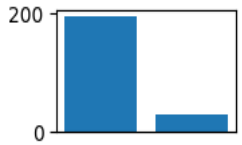
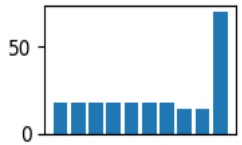
age	gender	pid	diagnosis	present	gxd	age_bin
65	F	patient 1	Coronary artery disease	YES	F_Coronary artery disease	60-70
65	F	patient 1	arterial hypertension	YES	F_arterial hypertension	60-70
65	F	patient 1	Acute MI	NO	F_Acute MI	60-70
65	F	patient 1	Transient ischemic attack	NO	F_Transient ischemic attack	60-70
65	F	patient 1	left ventricular hypertrophy	NO	F_left ventricular hypertrophy	60-70
65	F	patient 1	Earlier MI	NO	F_Earlier MI	60-70
65	F	patient 1	Sinus node dysfunction	NO	F_Sinus node dysfunction	60-70
59	M	patient 2	Coronary artery disease	NO	M_Coronary artery disease	50-60
59	M	patient 2	arterial hypertension	NO	M_arterial hypertension	50-60
59	M	patient 2	Acute MI	YES	M_Acute MI	50-60
59	M	patient 2	Transient ischemic attack	NO	M_Transient ischemic attack	50-60
59	M	patient 2	left ventricular hypertrophy	NO	M_left ventricular hypertrophy	50-60
59	M	patient 2	Earlier MI	NO	M_Earlier MI	50-60
59	M	patient 2	Sinus node dysfunction	NO	M_Sinus node dysfunction	50-60
80	M	patient 3	Coronary artery disease	NO	M_Coronary artery disease	70-80
80	M	patient 3	arterial hypertension	NO	M_arterial hypertension	70-80
80	M	patient 3	Acute MI	NO	M_Acute MI	70-80
80	M	patient 3	Transient ischemic attack	YES	M_Transient ischemic attack	70-80
80	M	patient 3	left ventricular hypertrophy	NO	M_left ventricular hypertrophy	70-80
80	M	patient 3	Earlier MI	NO	M_Earlier MI	70-80
80	M	patient 3	Sinus node dysfunction	NO	M_Sinus node dysfunction	70-80
51	F	patient 4	Coronary artery disease	NO	F_Coronary artery disease	50-60
51	F	patient 4	arterial hypertension	NO	F_arterial hypertension	50-60
51	F	patient 4	Acute MI	NO	F_Acute MI	50-60
51	F	patient 4	Transient ischemic attack	NO	F_Transient ischemic attack	50-60
51	F	patient 4	left ventricular hypertrophy	NO	F_left ventricular hypertrophy	50-60
51	F	patient 4	Earlier MI	NO	F_Earlier MI	50-60

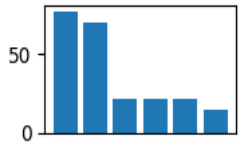
51	F	patient 4	Sinus node dysfunction	NO	F_Sinus node dysfunction	50-60
68	M	patient 5	Coronary artery disease	NO	M_Coronary artery disease	60-70
68	M	patient 5	arterial hypertension	NO	M_arterial hypertension	60-70

Features View

Stats

feature	missing_values	min	max	mean	std	1st_percentile	2nd_percentile	3rd_percentile	type	values	distribution	sensitive	target
age	0	18.0	80.0	56.1875	14.1769 9219239 816	50.5	59.0	66.0	integer	18, 19, 38, 39, 40, 41, 45, 49, 51, 52, 53, 56, 57, 58, 59, 60, 62, 64, 65, 66, 68, 70, 73, 74, 80		False	False
gender	0								binary	F, M		True	False
pid	0								categorical	patient 1, patient 10, patient 11, patient 12, patient 13, patient 14, patient 15, patient 16, patient 17, patient 18, patient 19, patient 2, patient ...		False	False

feature	missing_values	min	max	mean	std	1st_percentile	2nd_percentile	3rd_percentile	type	values	distribution	sensitive	target
diagnosis	0								categorical	Acute MI, Coronary artery disease, Earlier MI, Sinus node dysfunction, Transient ischemic attack, arterial hypertension, left ventricular hypertrophy		True	False
present	0								binary	NO, YES		False	True
gxd	0								categorical	F_Acute MI, F_Coronary artery disease, F_Earlier MI, F_Sinus node dysfunction, F_Transient ischemic attack, F_arterial hypertension, F_left ventricula...		True	False

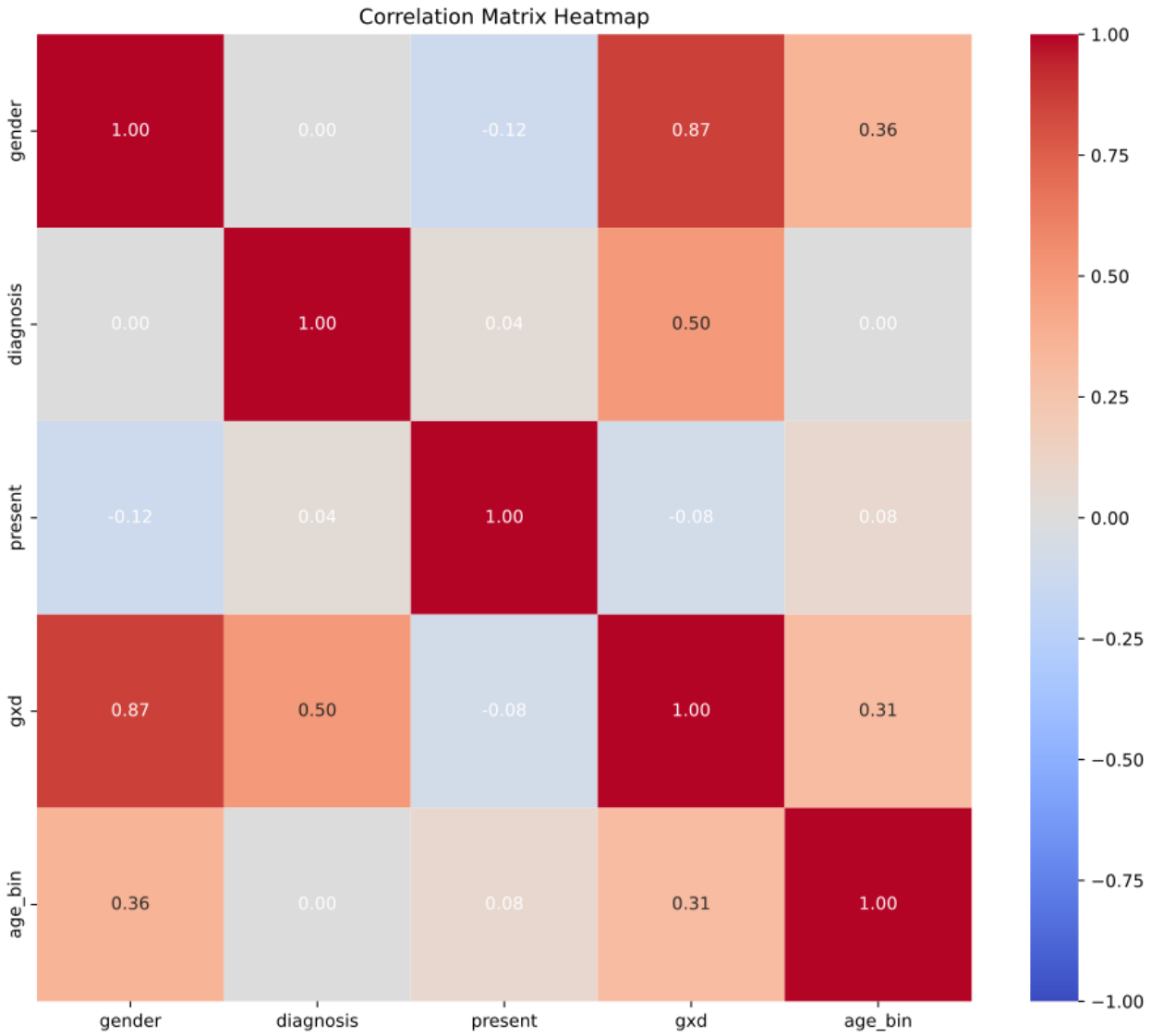
feature	missing_values	min	max	mean	std	1st_percentile	2nd_percentile	3rd_percentile	type	values	distribution	sensitive	target
age_bin	0								categorical	0-20, 30-40, 40-50, 50-60, 60-70, 70-80		True	False

Features

Field Name	drop	sensitive	target
age	✔	✘	✘
gender	✘	✔	✘
pid	✔	✘	✘
diagnosis	✘	✔	✘
present	✘	✘	✔
gxd	✘	✔	✘
age_bin	✘	✔	✘

Proxies

Correlation Matrix



Suggested Proxies

Gender

	diagnosis	gxd	age_bin
correlation	0.00%	86.66%	35.75%
suggested_proxy	✗	✓	✗

Diagnosis

	gender	gxd	age_bin
correlation	0.00%	49.91%	0.00%
suggested_proxy	✗	✗	✗

Gxd

	gender	diagnosis	age_bin
correlation	86.66%	49.91%	30.98%
suggested_proxy	✓	✗	✗

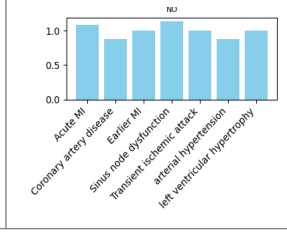
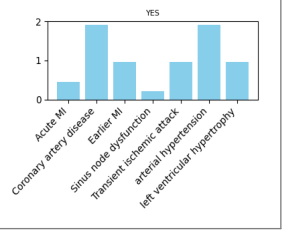
Age_bin

	gender	diagnosis	gxd
correlation	35.75%	0.00%	30.98%
suggested_proxy	✗	✗	✗

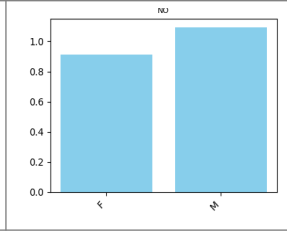
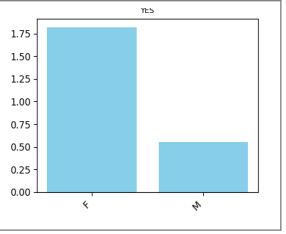
Detection

Selected Metrics

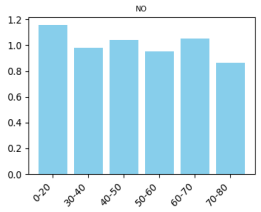
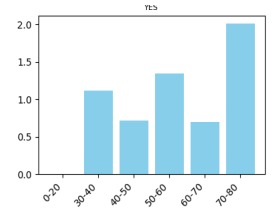
DisparateImpact

	NO	YES
diagnosis		

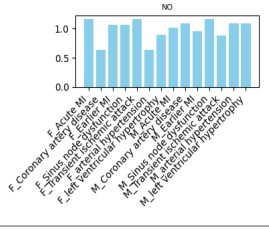
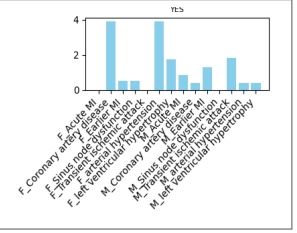
DisparateImpact

	NO	YES
gender		

DisparateImpact

	NO	YES
age_bin		

DisparateImpact

	NO	YES
gxd		

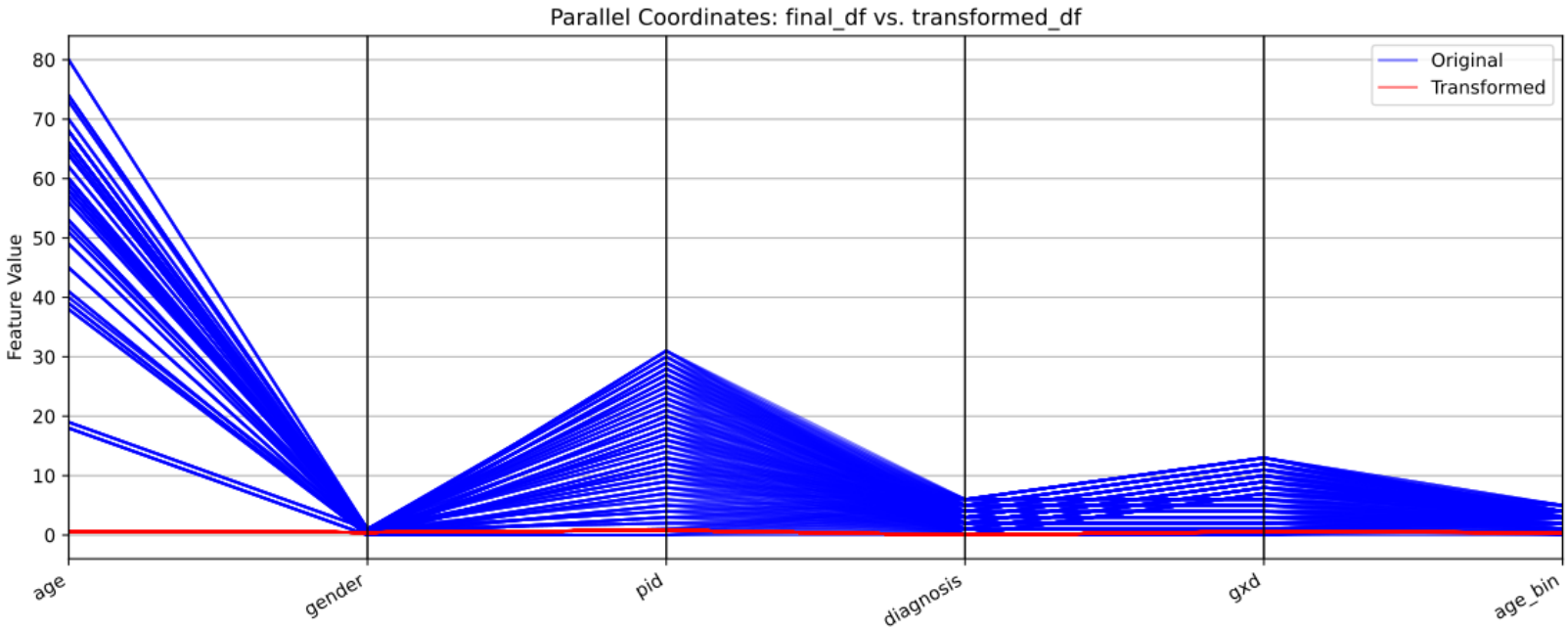
Data Mitigation

Preprocessing

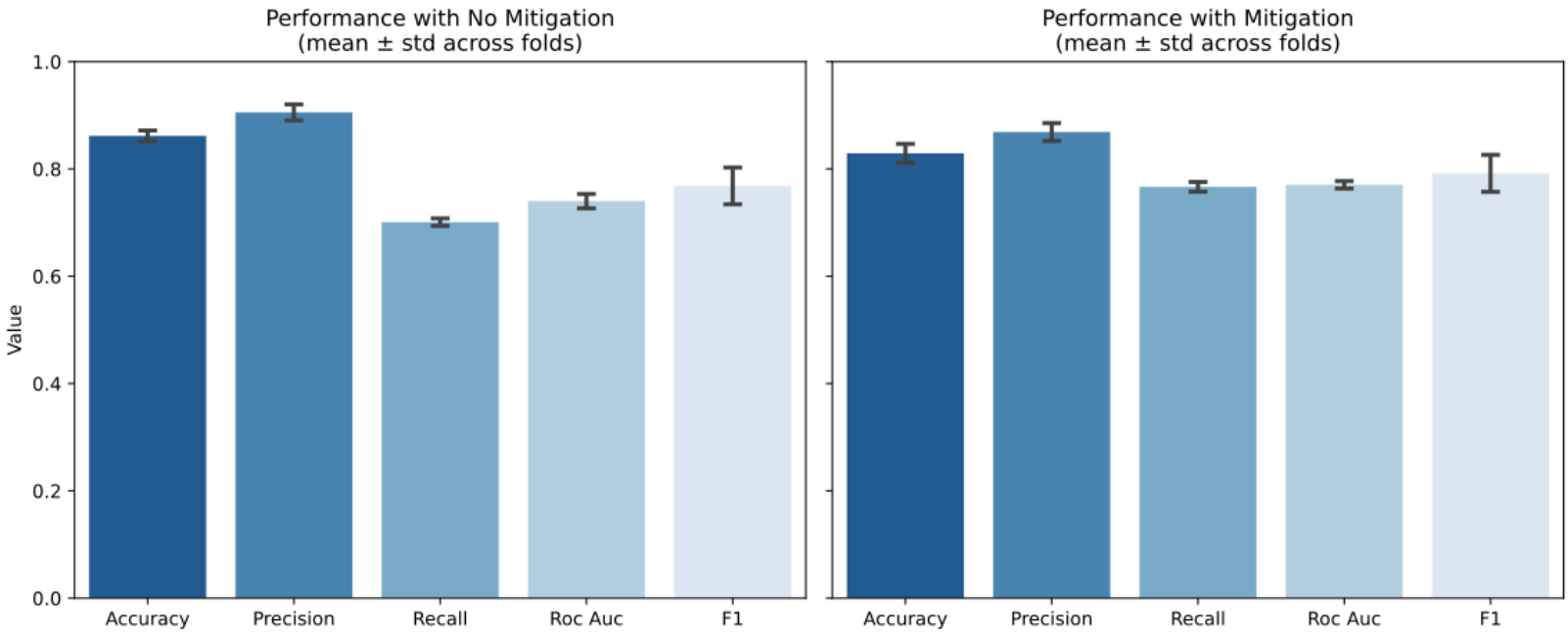
$algorithm	k	Ax	Ay	Az	random_state
LearnedFairRepresentations	5	1.00%	1	50	42

Data Mitigation Results

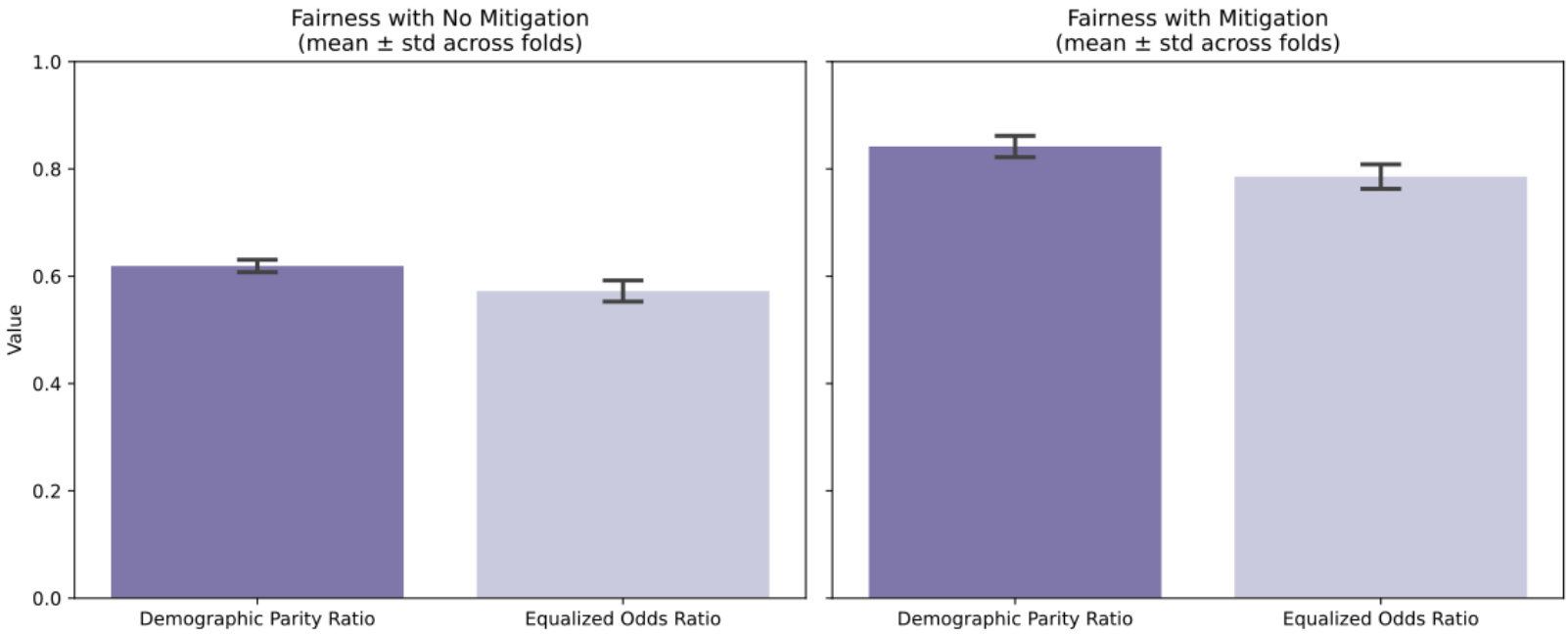
Preprocessing Plot



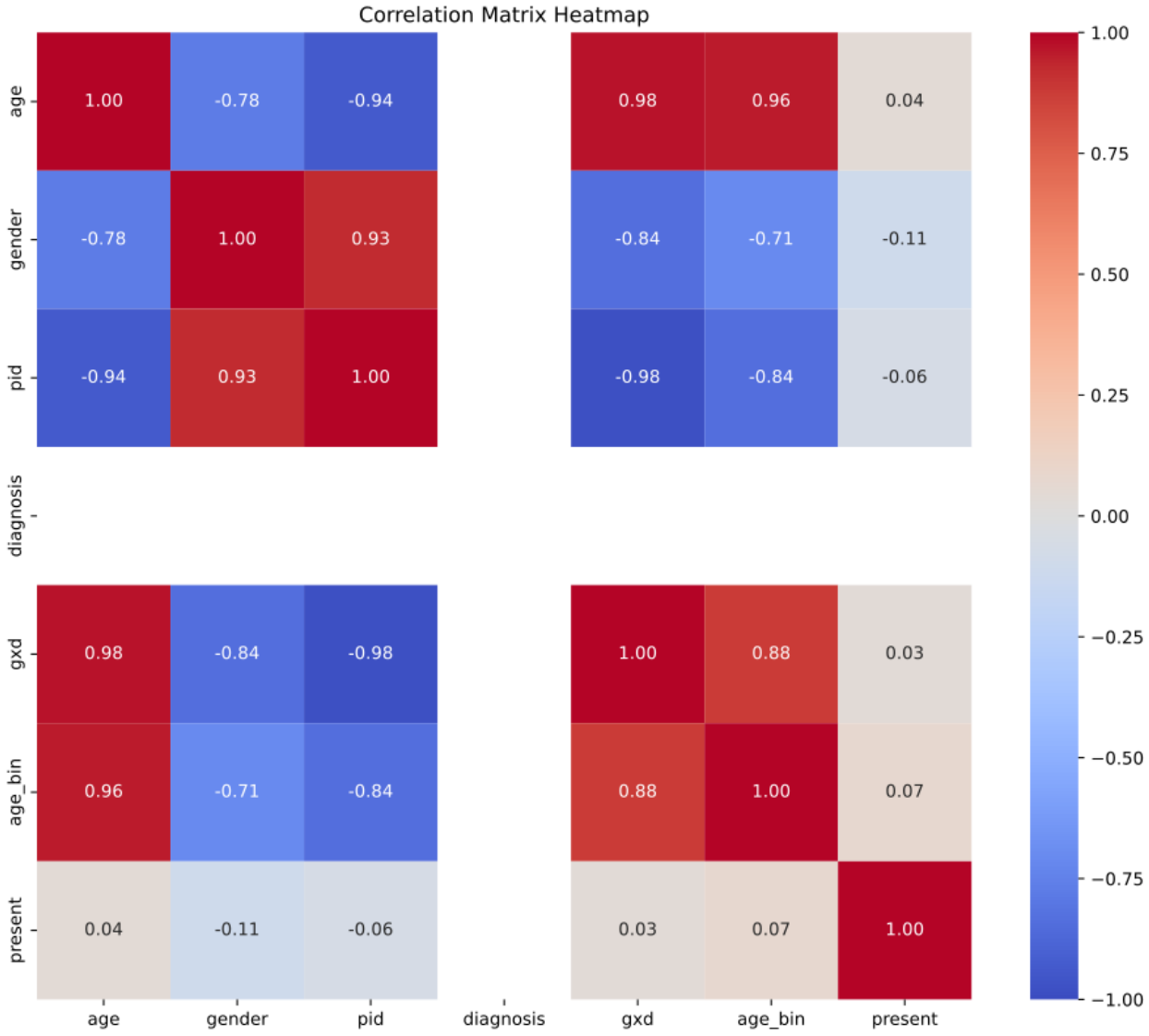
Performance Plot



Fairness Plot



Correlation Matrix



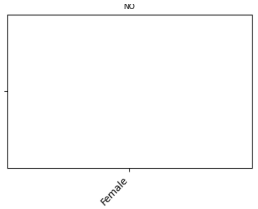
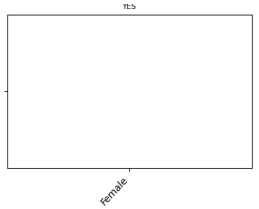
Dataset Head

age	gender	pid	diagnosis	gxd	age_bin	present
0.6656144576408652	0.440885530519474	0.8159540191302392	Female	0.5901509858583455	0.3973430815355205	YES
0.6674947201146189	0.4396370806883082	0.8144637072858607	Female	0.5957376891425883	0.3991652959230937	YES
0.6651280124904642	0.4411937076074654	0.8163307580420114	Female	0.5887237746838385	0.3968619567993906	NO
0.6670351663698108	0.4399511176413098	0.8148332449485521	Female	0.5943613129523602	0.398725599374276	NO
0.6679467387210798	0.4393221305561976	0.8140966349843103	Female	0.5970989743718554	0.3995940540736226	NO
0.6660947011823399	0.4405755624102876	0.81557867019384	Female	0.5915669889472331	0.3978142470881524	NO
0.6665683849037664	0.4402640185851929	0.8152049880811413	Female	0.5929707613899383	0.3982751721606399	NO
0.6679068518282497	0.4432073747020852	0.8156435454024762	Female	0.5951201274097596	0.3987355838017375	NO
0.6696961894713958	0.4418426218851476	0.8141147291999432	Female	0.6007070916518801	0.4003806206713028	NO
0.6674345875839665	0.4435450982353459	0.8160343570870511	Female	0.5936715091899003	0.3982910776675689	YES
0.6692643200333436	0.4421853154087954	0.8144912189259295	Female	0.5993432210841296	0.399989506621035	NO
0.6701174015690039	0.4414994004273443	0.8137424721570725	Female	0.6020477713598212	0.400758224787363	NO
0.6683694126218284	0.442867978570155	0.8152558952092189	Female	0.5965489014995465	0.3991669068511972	NO
0.6688219865700753	0.4425271964397678	0.8148716946769526	Female	0.5979568902737703	0.3995849138951526	NO
0.6665330304476218	0.4454810050517778	0.8174563905267758	Female	0.5898692539203921	0.3981810064664225	NO
0.667898010232496	0.4445173631651584	0.8163293109816846	Female	0.5940630717937521	0.3994585610409569	NO
0.6661799288337834	0.4457185769350092	0.8177413161552085	Female	0.5887977393392971	0.3978447005074086	NO
0.6675641251986087	0.4447600664520078	0.8166089570349214	Female	0.5930292405579421	0.3991494579507162	YES
0.6682268029247254	0.4442736424851572	0.8160512655535902	Female	0.5950865407968112	0.3997607182778891	NO
0.666881516123351	0.4452419931659411	0.8171726106352918	Female	0.5909319463103774	0.3985106178605782	NO
0.6672252553769191	0.4450016452194802	0.816890094384876	Female	0.5919854064474618	0.3988334575472032	NO
0.6612657917652359	0.4492143724192117	0.8214093000270951	Female	0.5756612961823144	0.3922610457020306	NO
0.6634175785477603	0.4478898417446915	0.8197362199630756	Female	0.5821115258400876	0.3943966861841366	NO
0.6607099791327403	0.4495333009955282	0.8218283939449856	Female	0.5740205376794945	0.3916969949482551	NO
0.6628913895375345	0.4482277455045571	0.8201532076216239	Female	0.5805189307949876	0.3938816872161041	NO
0.6639351280027314	0.4475479151423914	0.8193207323334956	Female	0.5836884024990862	0.3948984876528292	NO
0.6618149889641731	0.4488902933775382	0.8209901400790522	Female	0.5772923173210086	0.3928134728494165	NO

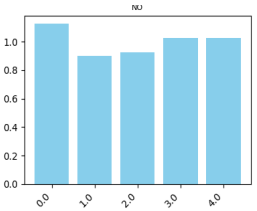
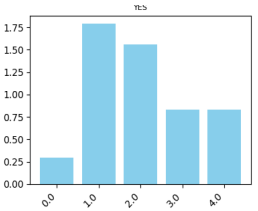
0.6623570281504837	0.4485613257244382	0.8205713110898308	Female	0.5789120365129318	0.3933538246300059	NO
0.6655682479672927	0.4470777974857688	0.8184602415916212	Female	0.5872393484026208	0.3967448216594624	NO
0.6671008582334605	0.4459838368534048	0.817185444656433	Female	0.5919855249379457	0.398183207079514	NO

Selected Metrics

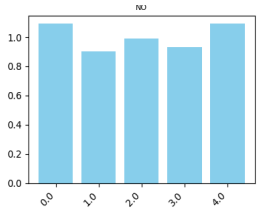
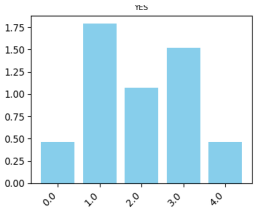
DisparateImpact

	NO	YES
diagnosis		

DisparateImpact

	NO	YES
age_bin		

DisparateImpact

	NO	YES
gxd		

DisparateImpact

	NO	YES
gender	